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Innovative Well Design For CO₂ Injection Into Depleted Gas Fields - Aramis Phase 1 Development Example

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Abstract

When injected into depleted gas fields, CO₂ is usually kept in dense phase for pipeline transportation. Standard CO₂ injectors design uses narrow tubing diameters to create friction and prevent CO₂ from free-falling into the well due to its column weight. This results in very narrow well operating envelopes which hinder pipeline operability (limited CO₂ compressibility when transported in dense phase) and necessitates regular workovers to maintain injection potential as reservoir pressure increases over time.

To improve operability of the entire carbon transportation and injection chain, the operating envelope of injection wells had to be widened. Well design had also to be compatible with low operating expenses and as such should avoid workover needs when reservoir pressure increases. The proposed new design is based on a controlled dual phase injection coupled with the usage of retrievable downhole chokes. The impact on wells' operating envelopes, architecture, equipment, and workover requirements has been assessed through numerical simulations and specific material qualification tests under operating conditions never seen in hydrocarbon wells.

Applied to Aramis Phase 1 development, the new design has widened the well operating envelopes by a factor 2 and has allowed adjusting the operating envelopes to the increasing reservoir pressure deriving from the CO₂ injection. It results in an enhanced pipeline and wells operability, thanks to the optimization of the combinations of larger individual wells operating envelopes. More specifically, the new design improves the wells redundancy (ensuring the contractual field target injection rate can still be achieved if one well is down), broadens the range of continuous feasible injection rates at the field level, and improves the management of injectivity uncertainty. Additionally, the new design is expected to avoid performing an average of 4 costly workovers at field level during the full injection lifespan. From a material point of view, simulations and equipment tests have shown no significant issue in terms of erosion, mechanical loads and corrosion.

The proposed new well design has achieved a global optimum across the entire CO₂ chain, from transportation to injection, enhancing asset profitability, operability, and well design, as well as reservoir management. The innovative well design implemented in the ARAMIS project has been pivotal in unlocking this project and is expected to extend the applicability of CCS injection into depleted gas fields.

Introduction

In collaboration with various local partners and initiatives, the ARAMIS project [1] aims to investigate the development of CO₂ transport facilities to provide access to offshore CO₂ storage for hard-to-abate industries, like cement plants or steel industries. The facilities will include, amongst others:

- An onshore CO₂ collection hub (number 3 on Figure 1 below) located in the Port of Rotterdam. The CO₂ from potential customers can be transported to the hub via ships (coaster ships and river barges) as well as through onshore pipelines in gaseous phase (number 2 on Figure 1). The onshore CO₂ collection hub will consist of a compressor station and a shipping terminal with temporary cryogenic storage facilities for the liquid CO₂ arriving by ship.
- An offshore trunkline, 32"-Carbon Steel pipeline that will transport the CO₂ from the collection hub to the distribution hub platform, where the CO₂ is distributed to the various injection platforms (number 4).
- Injection platforms to inject CO₂ into depleted offshore gas fields, located 3-4 km below the seabed (number 5).

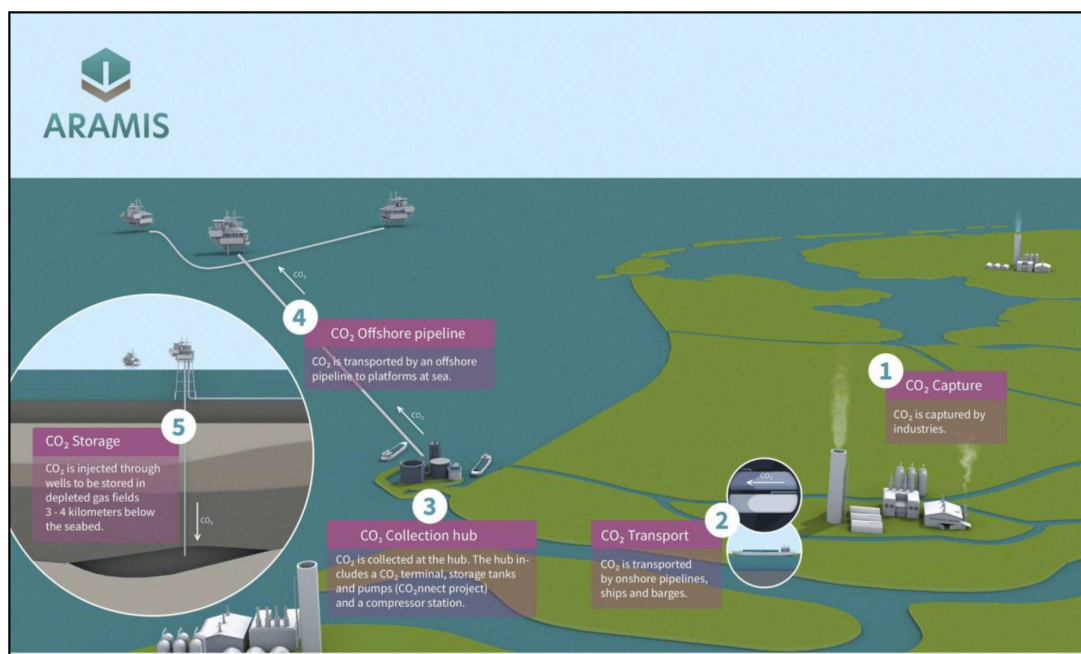


Figure 1—ARAMIS project overview

The capture of CO₂ (number 1) is outside the scope of ARAMIS.

Due to the length of the main trunkline (~200 km), CO₂ is transported in dense phase. The aim is to reduce friction pressure losses and avoid excessive high pressure at trunkline departure. It induces a limited flexibility of the global system in terms of rate, due to the limited compressibility of dense CO₂. Hence, to avoid important variations of pressure in the trunkline, injected rates through wells must be very close to inlet trunkline rates.

Considering an available pressure upstream of the wellhead chokes close to 130-150b, injection wells are facing many challenges. Two main ones are:

- The control of injection rate at early life. Reservoirs are located at a depth of around 3750 mTVD/ ss and will have a pressure close to 20b at the start of injection. Assuming a pressure upstream of the wellhead chokes close to 130-150b and a CO₂ density similar to liquid water (dense phase), it

leads to a bottom hole pressure of approximately 500b if friction losses are neglected. Overpressure on reservoir would then be around 480b which is far too high to control the injection rate.

- Getting enough injection potential at the end of injection life. Indeed, reservoir pressure will increase over time due to injected CO₂. Ideally, one will repressurize the reservoir until its initial value (~ 440 b) and one will potentially lack of reservoir overpressure to inject target rate.

For the well design, to deal with these 2 opposite objectives (decrease of the injection potential at early life while maximization of it at late life), the industry was using a traditional design based on monophasic injection all along the well, tubing friction and late-life workovers. Our innovative design is based on dual phase flow injection coupled to the usage of retrievable downhole chokes. It presents many advantages that will be detailed in the next sections.

Traditional design (monophasic injection all along the well based on tubing friction)

Principle and resulting well operating envelope

The idea is to reduce the injection potential thanks to tubing friction at early life, and perform workovers to install bigger tubing's when the injection potential becomes too low. The resulting well operating envelope for one well is illustrated on [Figure 2](#) hereafter.

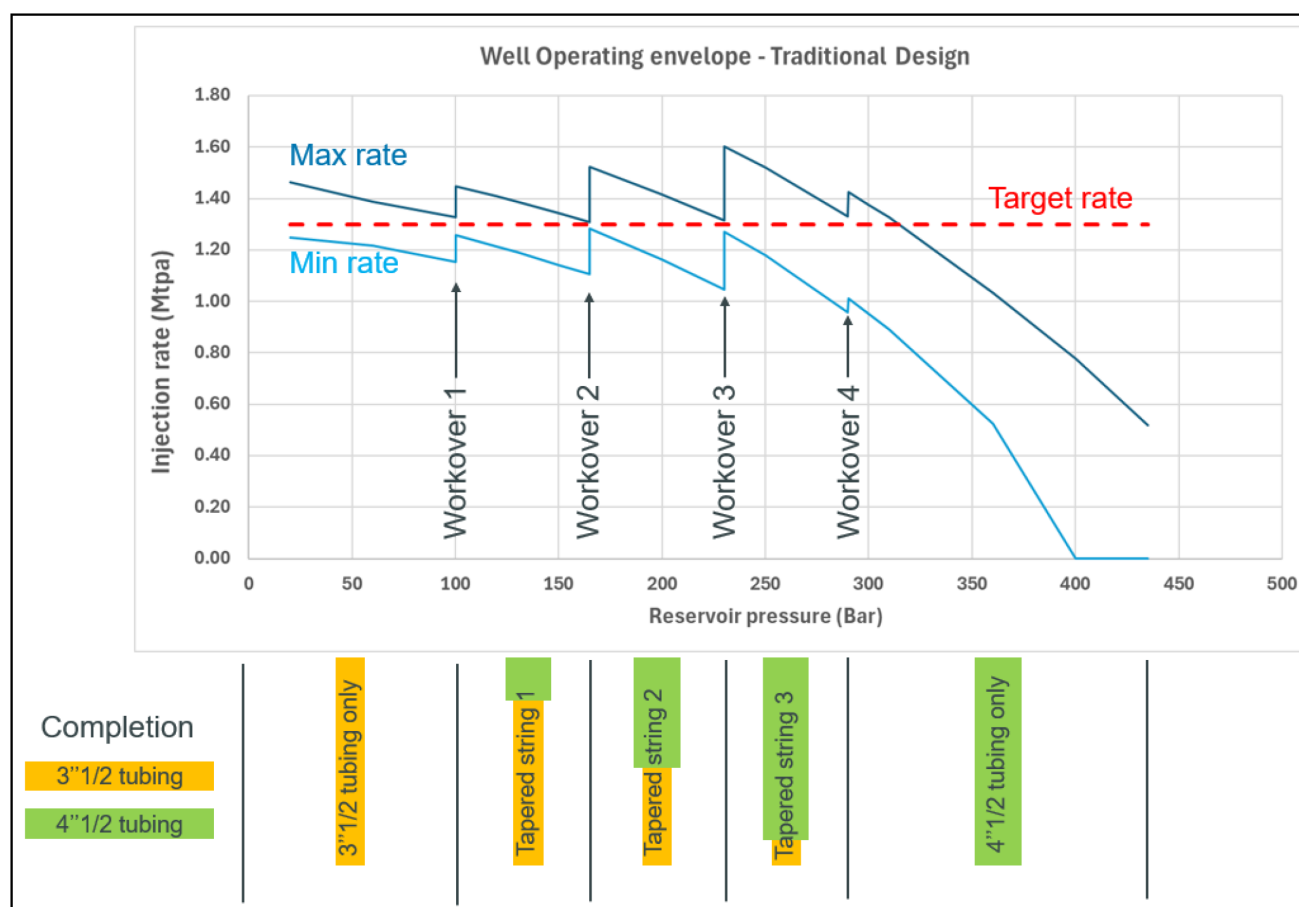


Figure 2—Traditional design - well operating envelope

Main advantage of such solution is that liquid being monophasic into the full well, no subzero temperatures are observed into the well in steady-state injection.

As can be seen on [Figure 2](#), such solution has unfortunately many drawbacks among which:

- A limited well operating envelope leading to more opening and closing of the well if CO₂ delivery deviates from what is planned.
- The need of many workovers to cover the full injection lifetime (installation of bigger tubing's as reservoir repressurizes).
- The poor resilience of the system to injectivity uncertainty.

To reduce the drawbacks of the traditional design, a new well design has been proposed for the phase 1 of the ARAMIS project. It is detailed in the next section.

Innovative well design for CCS injectors

Principle and resulting well operating envelope

The new well design is based on two key elements:

- A controlled dual phase injection to enlarge the well operating envelopes. The dual phase injection is controlled to avoid generating too much cold into the well due to the vaporization of liquid CO₂ (latent heat and Joule Thomson effect). As can be seen on [Figure 3](#) for the illustrative well, a wellhead choking down to -5°C allows to enlarge the well operating envelope by a factor 2. It is enlarged by a factor 5 when choked down until -10°C.
- The usage of retrievable downhole chokes to avoid workovers as reservoir repressurized. It also enables to adjust well potential to injectivity uncertainty. As illustrated on [Figure 4](#), installing retrievable downhole chokes (DHC) in a 4 ½ tubing allows to have operating envelopes varying between the one of a 3 ½ tubing (with 3 DHCs of size 1 in studied case) and the one of a 4 ½ tubing (without DHCs). As such, it provides a lot of flexibility in terms of well operating envelopes as they can be moved up or down depending on the needs.

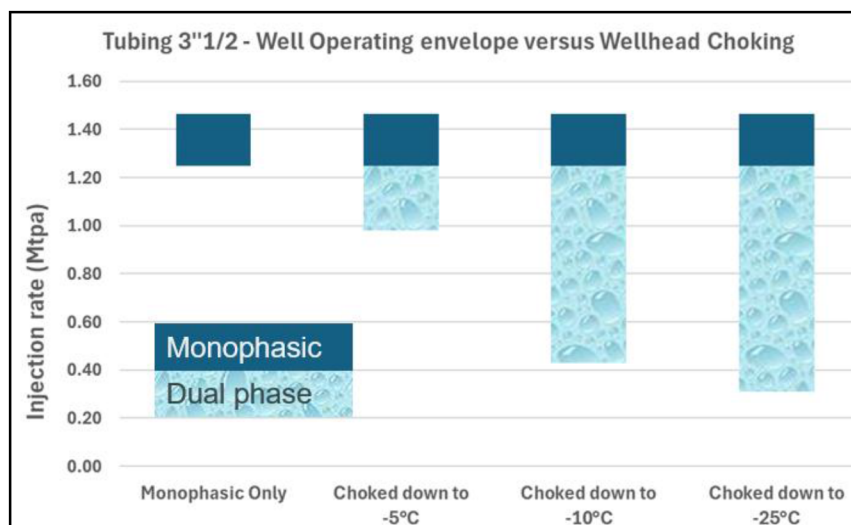


Figure 3—Impact of wellhead choking on Well Operating envelope (start of injection)

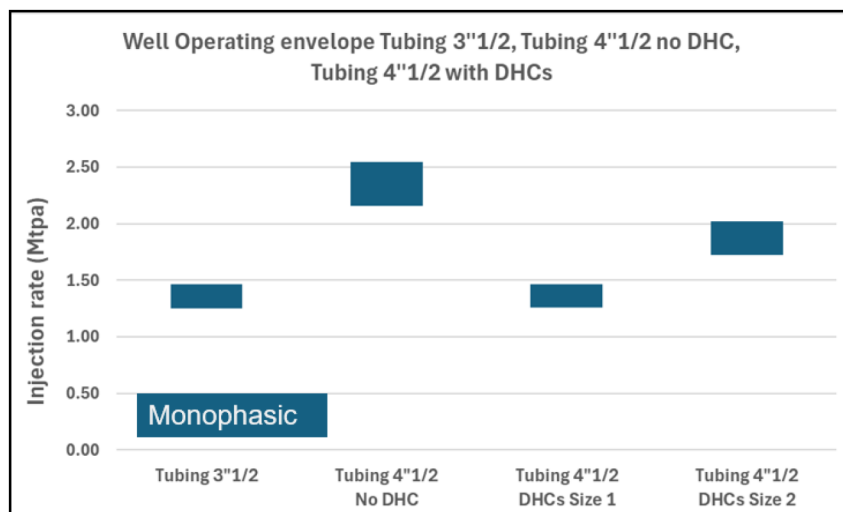


Figure 4—Impact of tubing size and downhole chokes on Well Operating Envelope (start of injection)

Figure 5 and Figure 6 illustrate the full potential of the innovative design. By comparing them with Figure 2, one can see that even when wellhead choking is limited to -5°C only:

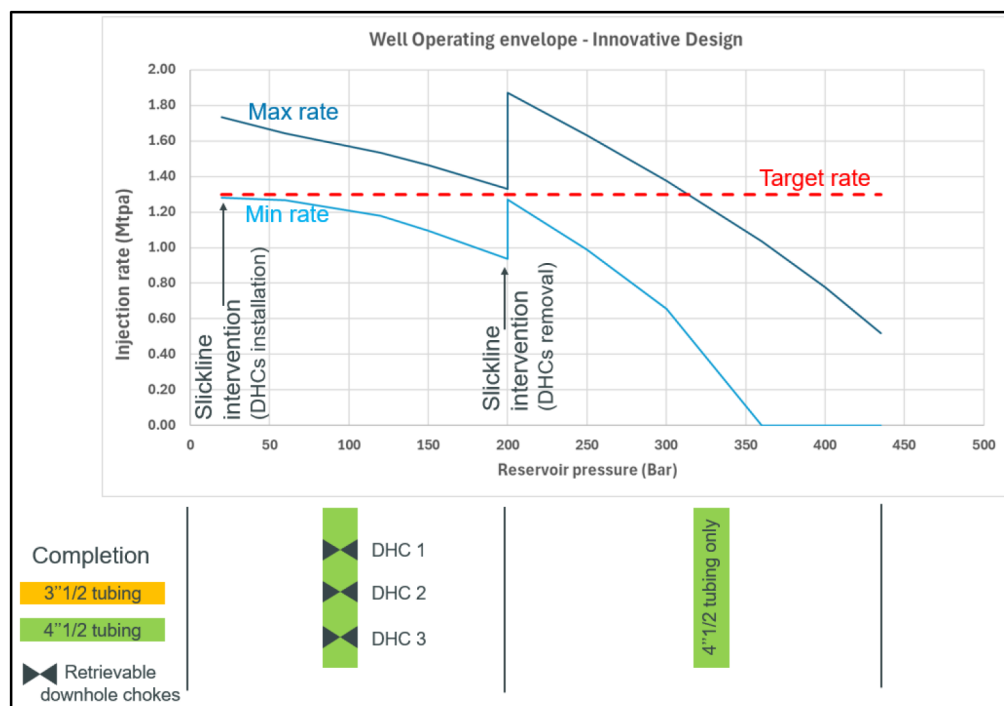


Figure 5—Innovative well design – well operating envelope with dual phase injection limited to -5°C

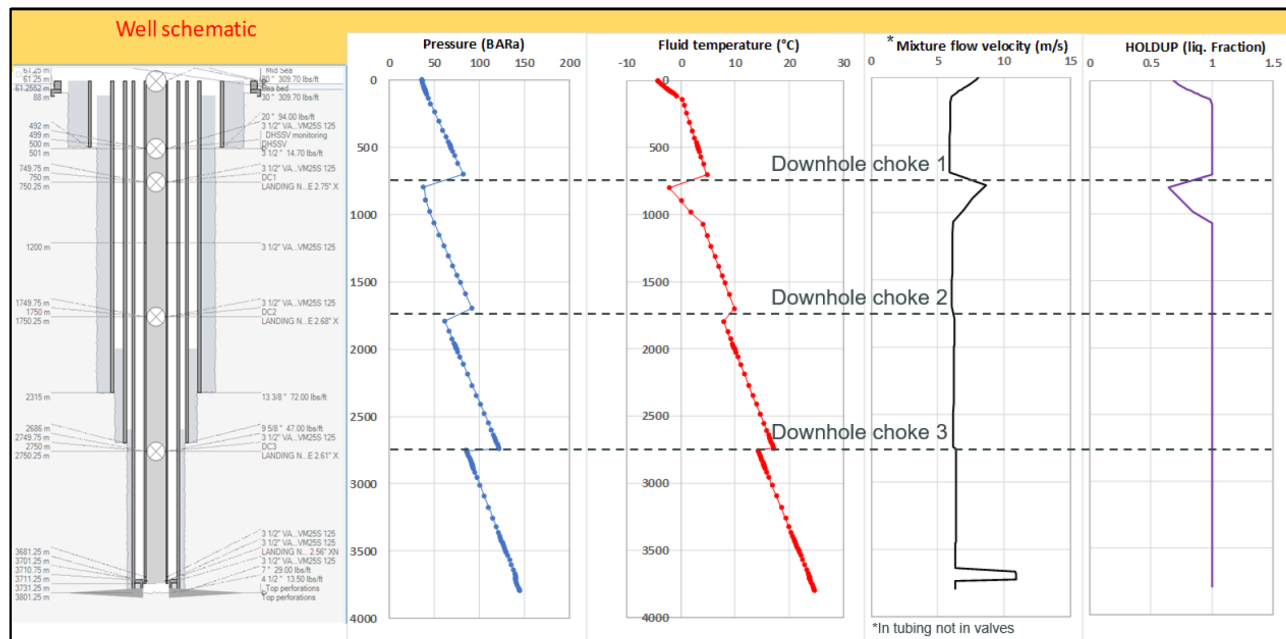


Figure 6—Innovative well design – initial well profiles with dual phase injection limited to -5°C

- The well operating envelope is enlarged by a factor 2;
- Only two slickline interventions are needed to cover the full injection lifetime as big tubing is installed from the beginning (no need for workovers to change tubing's) and downhole chokes can be changed with simple slickline interventions;
- The system is resilient to injectivity uncertainty as downhole chokes are retrievable and can be adjusted when needed.

Remarks:

- Here below, only the case with dual phase injection limited to -5°C is presented. In case people are ready to operate the wells in dual phase injection down to -10°C or lower, the system is then even more flexible, and no downhole chokes are needed.
- Usage of DHCs in CO₂ injectors has been considered in the past [2], [3]. The aim of DHCs usage was nonetheless different as it was to maintain the CO₂ in dense phase in the full well rather than switching the dual phase operating envelope up or down, and decide where cold is generated into the well.

Completion design considerations for sub-zero dual phase wells

Number of Chokes versus Delta-P and velocity

One of the first design parameters to take into consideration was the magnitude of the Delta-P to be achieved globally along the flow path in the well by means of the Down Hole Chokes. Depending on the scenario (skin values, well size, ...), the DHCs were required to provide between 200 and 300 bars of cumulative delta-P. The design consists then in balancing the number of chokes versus the delta-P that each choke would have to generate: the greater the number of chokes the less the delta-P across each choke. A low delta-P is favorable in terms of erosional and other mechanical impacts on the choke. However, some practical aspects also need to be taken into consideration in the design: from a well intervention perspective for example, having fewer chokes in the well will reduce the number of slickline runs to retrieve or change chokes. Here an optimum was selected by limiting the number of DHCs to a maximum of 3 and the Delta-P on a choke

to approximately 100 bars. Additionally, an adjustable choke is placed next to the wing valve on the X-mass tree.

This constraint of max delta-P is also associated with the maximum velocity that will be seen by the choke insert. An upper limit had been initially set to 150 m/s based on erosion tests performed with air and abrasives by choke manufacturers, using different inserts materials. With 3 DHCs and +/-100 bar max Delta-P per choke, the velocity limitation can be honored while satisfying the total delta-P requirement.

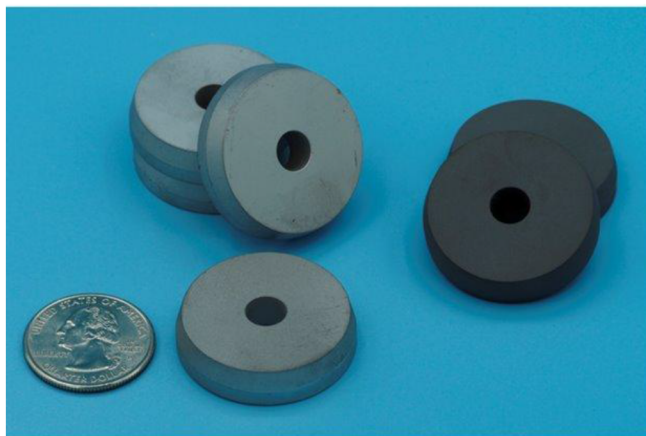


Figure 7—Various choke inserts (Tungsten Carbide & PDC).



Figure 8—Choke holder (with insert shown in yellow). To be connected to a lock mandrel & installed in a nipple profile.

Erosion verification

Currently (2025), only a few CO₂ flow loops are available to physically test equipment. They can usually accommodate moderate CO₂ circulation rates (approx. 0.3 Mtpa max), they can seldom accommodate dual-phase and don't have the possibility to incorporate solids for erosion evaluation. Therefore, CFD (Computational Fluid Dynamics) has to be used for this purpose. A 3D geometry of the full choke assembly (i.e. including the lock mandrel and insert holder) has been plugged in a CFD model and various rates and solid content have been simulated. The CFD model had been previously calibrated for the erosion aspect

using actual physical tests (with solid laden water) and also matched in terms of Delta-P & Delta-T using solutions obtained from our multiphase flow simulations software.

Sensitivities were done assuming particle size up to 50 μm , solid loading up to 0.3 lb/1000 bbl of liq CO_2 (0.1 lb/mm³scf) and using quartz (SiO_2) particles as abrasive material. This is deemed to represent a more severe condition than what will be occurring in the reality due to the fact that the specifications of the injected CO_2 will be extremely stringent (especially CO_2 from cryogenic origin). Filters (<50 μm) will be installed upstream of the wells (and also upstream of the pipeline) and, being injector wells, no solids are expected from the reservoir (typically quartz) as usually used for completion items qualification for producer wells. The most likely source of solids would be iron oxides particles from the pipeline (original mill scale or pipe initial corrosion) which is much less abrasive than quartz.

Note, for comparison, that the API-19ICV [4] for example, considers for erosion tests using particle sizes with D_{50} of 10 to 30 μm (much smaller, hence less abrasive). Also in this API, the Standard Erosive Service for a producer well considers a solid loading of 1 lb/1000 bbl and the industry generally uses 10% (or less) solid concentration for injectors than for producers (hence 0.1 lb/1000 bbl).

Erosion levels obtained from the CFD simulations were deemed acceptable, well below the set criteria of no increase of choke diameter superior to 10% over the life of the choke, with less than 1% of the surface area of the choke showing erosion rates above 0.01 mm/yr with Tungsten Carbide inserts.

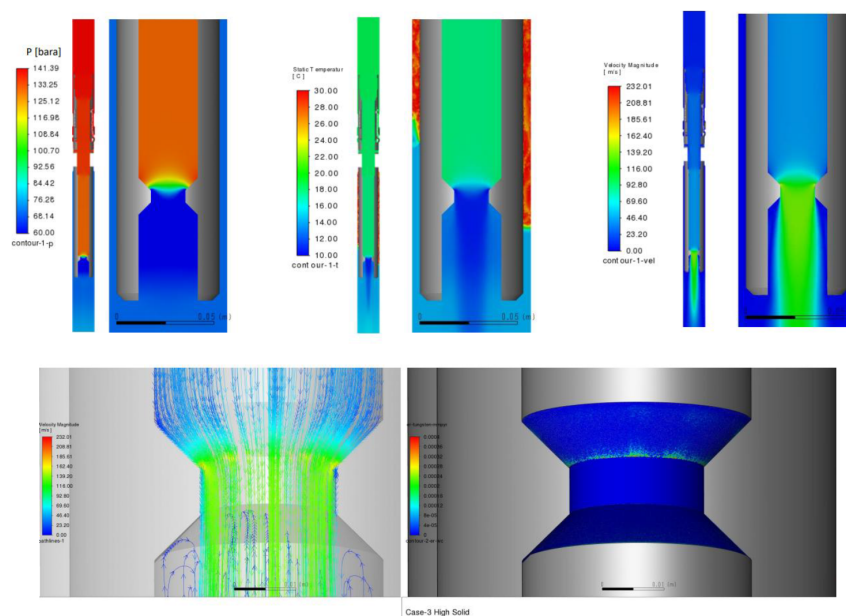


Figure 9—CFD simulation results (P, T, Velocity & erosion) of CO_2 flow through a DHC

DHC positioning in the well

Another design criteria pertained to the relative position of the Down Hole Chokes in the wellbore.

First, an objective was having the CO_2 to enter the formation as a dense phase. As the CO_2 exits the DHCs in dual phase, it means that downstream of the lowermost choke, there needs to be sufficient distance (in TVD) to the reservoir to build back some hydrostatic pressure, thereby recompressing the CO_2 back to dense phase.

It was also desirable to have the CO_2 entering the DHC themselves in dense phase. Here again, the distance (in TVD) between two DHC must be sufficient to recompress the CO_2 back to dense phase.

Finally, while honoring the requirement for distance between chokes, the actual position can be adjusted in order to select where the cold created by the phase change downstream of the choke (maximum at outlet

of DHC) will occur, for example the first DHC was selected to be set approximately 250m below the DHSV (750mTVD/TH versus 500mTVD/TH).

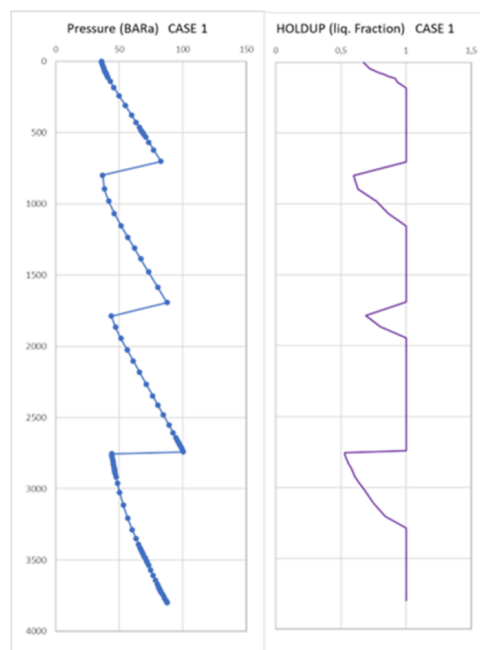


Figure 10—Pressure profile versus depth (highlighting the Delta-P across DHC and the liquid hold-up (<1 indicates dual-phase): The CO₂ is in dense phase before entering the DHC and the formation

Consequences of operating in dual phase

Impact on the outside of the production tubing. Operating a well in sub-zero dual-phase, means that, contrary to the cooling due to transient, which is intense but of short duration, the cooling associated with dual-phase can be of extended duration (potentially months or years) i.e. in a Pseudo-Steady-State. The consequence is that, radially, the cold front has plenty of time to propagate outwards, from the tubing to the outer casings, annular spaces, cement layers and formation.

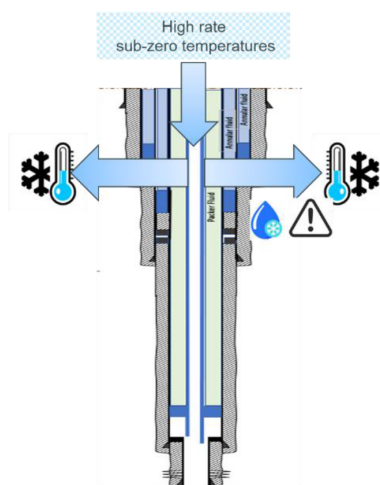


Figure 11—Outward extension of cold temperatures

It is therefore very important to scrutinize the freezing properties of the various fluids located in the annular spaces, i.e. whether they could freeze and if they freeze, whether their volume will increase (as

does freshwater ice). The consequence of freezing with volumetric expansion occurring in a closed annulus could be severe, potentially leading to mechanical failure of the tubulars [5].

These properties vary a lot depending on the nature of the fluid (Oil-based fluid, water-based fluids) and their composition.

In the case of new CCS injector wells to be drilled, the fluids subjected to negative temperatures can be designed to resist freezing, for instance by using suitable oil-based fluids for packer fluid or addition of antifreeze to water-based fluids such as cement spacers. However, in the case of re-usage of existing wells, where legacy fluids from the initial well construction remain, care must be taken to identify the various fluids present and assess their freezing properties.

Use of a Nitrogen cap in the first annulus. Due to the natural geothermal gradient, the deeper sections of the well are the warmest and the shallowest portion of the well (close to surface) is the least heated by the formation. In dual-phase operation, the surface choke (the one on the X-mass tree) is used to create the first phase change along the injection path, so negative temperatures in the tubing start right from the X-mass tree and extend several hundred meters down, thereby cooling the section of the well that is the least warmed up by the geothermal gradient. Since the B and C annuli are usually not cemented to surface, the upper section of the well is also where fluids are likely to be present in the annular spaces and therefore where the risk associated with freezing is the greatest (presence of liquid & presence of cold).

To mitigate this risk, a thermal insulation (in the form of a gaseous nitrogen cap) is placed in the A-annulus above the packer fluid.

With the moderate dual-phase selected in the project (i.e. not operating below -5°C in PSS), a nitrogen blanket extending 500m from surface was sufficient to ensure that, in steady state, the annular spaces in the upper section of the well would remain at positive temperatures. Below 500m the geothermal gradient is now sufficient to ensure positive temperature in annular spaces without insulation.

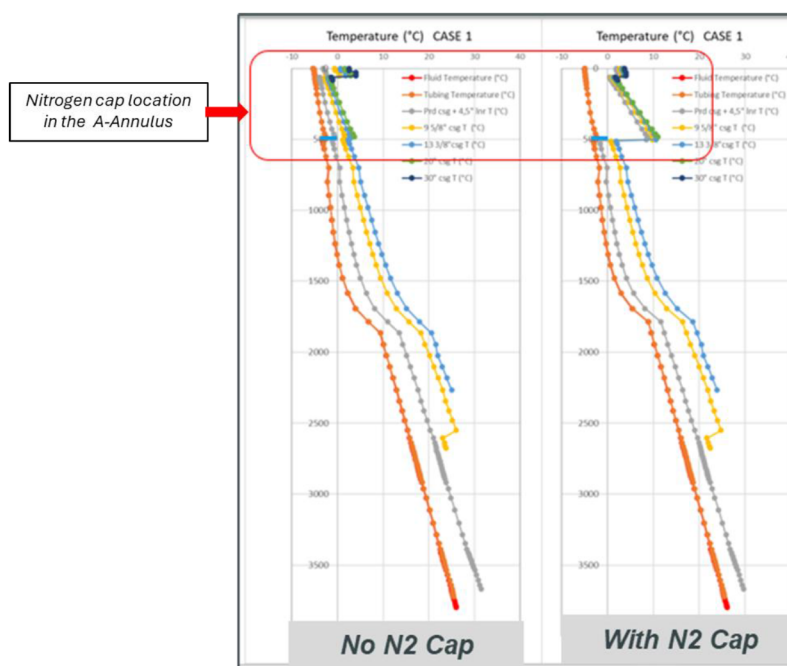


Figure 12—Comparison of the various casing strings temperature with and without N2 Cap

Note that the DHSV is placed below the nitrogen cap so as to not be isolated and benefit from the warmth of the formation.

From a thermal standpoint, the less the pressure of the nitrogen the better the insulation (so it is "failsafe" in case of N₂ leak). However, it is desirable to maintain a certain level of positive pressure on this N₂ cap (A-annulus Casing Head Pressure) for monitoring purposes of the A annulus (approx. 250psi after cooling).

Transients from Dual Phase injection.. When stopping the injection in a well, without a down-hole closure system, the fluid in the tubing will fall under its own weight, and depressurize, creating intense cold until the fluid column ultimately balances with the reservoir pressure (which at the early stages of injection can be extremely low in a depleted reservoir project). The transients generated upon shutting-in the injection while previously injecting in a sub-zero dual-phase mode will be colder than the transients generated from a single-phase injection situation. Therefore, these transients (from dual phase) are the ones to be scrutinized for establishing the worst-case scenario of low temperatures in the well and in the annular spaces. In transient simulations, flow models can predict the temperature versus time during the transient period for any given depth of the well and for each casing string. In order to evaluate the minimum temperature encountered and compare it to the freezing point of the annular fluids, a routine is created in the model to identify, during the entire transient duration, the lowest temperature anywhere, anytime, for each annulus.

In the example below (a side-track), the minimum temperature reached by the production casing during a shut-in transient is -7.9°C while the next casing (9 5/8) is at -1°C. The B-annulus fluid will therefore be designed to be able to not freeze at these temperatures. Note that this is a conservative approach since the dynamics of the freezing itself are not considered (the freezing of large mass of fluids will not be instantaneous whereas the transients are quite brief)

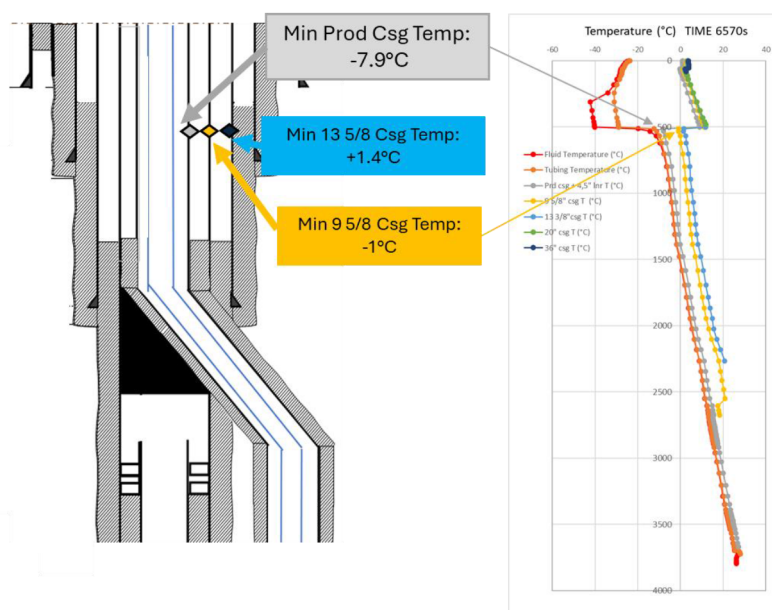


Figure 13—Minimum temperature on casings identified during a transient from shutting-in a dual phase

Impact on the inside of the production tubing. The selection of OCTG for CCS application in depleted reservoirs (i.e. undergoing intense cold transients) has led to defining new testing protocols to guarantee that the connections will maintain their gas-tight sealability [6], [7], [8]. These protocols include sustained negative temperature testing under various load (tension, compression) thereby catering for the cold dual-phase steady state conditions. The topic of resistance to brittleness transition (impact toughness) due to cold is also to be scrutinized for the tubing material selection, usually CRA materials offer good resistance.

However, in terms of resistance to corrosion, Dual-phase injection of anthropogenic CO₂ poses a particular challenge. Due to its anthropogenic origin, the CO₂ is not entirely pure and contains several contaminants, in particular SO_x & NO_x together with a minute fraction of water (few tens of ppm). As long

as any water (present as dissolved phase in the CO₂) stays in solution, not much issues exist but if water drops out of solution to become free water then it becomes available for chemical reactions. This water can react with the SO_x and NO_x in complex reactions that ultimately can generate very corrosive species such as Sulfuric or Nitric acids [9].

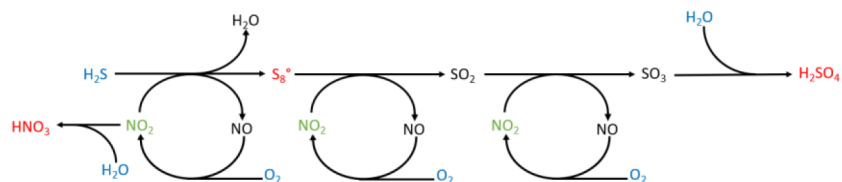


Figure 14—Reactional pathway focused on oxidation state of sulfur, from H₂S (-II) to sulfuric acid (+VI) [9]

The solubility of water depends essentially on P&T conditions, with solubility decreasing (i.e. risk increasing) with colder temperatures.

It is therefore important to check on the phase envelope if the operating point of the dual phase remains outside of the aqueous (free water) zone otherwise a potential situation of continuous release of acid could exist. By remaining at reasonable levels of cold for the dual-phase operating mode (in this case not below -5°C), the operating point offers sufficient margin to the aqueous zone to prevent such occurrence. The CO₂ phase envelope itself can have different shapes depending on its actual composition (contaminants type & concentrations), several hypotheses of composition were sensitized in the evaluation.

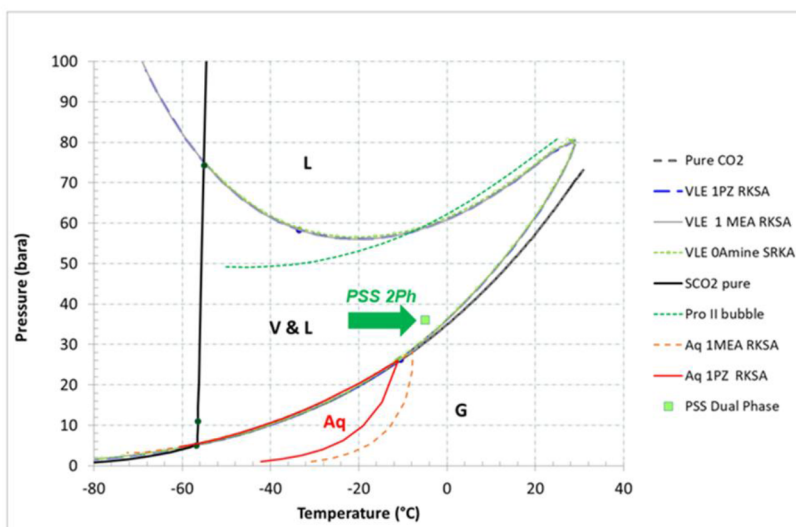


Figure 15—Dual-phase Pseudo Steady State operating point (green dot) versus aqueous zone (in red)

During the transient phases (associated with the shut-in of the well previously injecting and with the resumption of injection in a shut-in well), the temperature of the CO₂ can go much lower than in PSS (-30°C or less) and therefore can make an excursion into the aqueous zone. Although these transients are quite limited in time, the corrosion aspects have been investigated in the laboratory for different tubular materials [10]. The tests, performed on metallic coupons, involved several rapid decompressions of CO₂ containing impurities. They indeed demonstrated the release of corrosive species as predicted by the chemical models and showed that 13%Cr material would not be adequate to resist corrosion whereas 25%Cr CRA was able to resist.

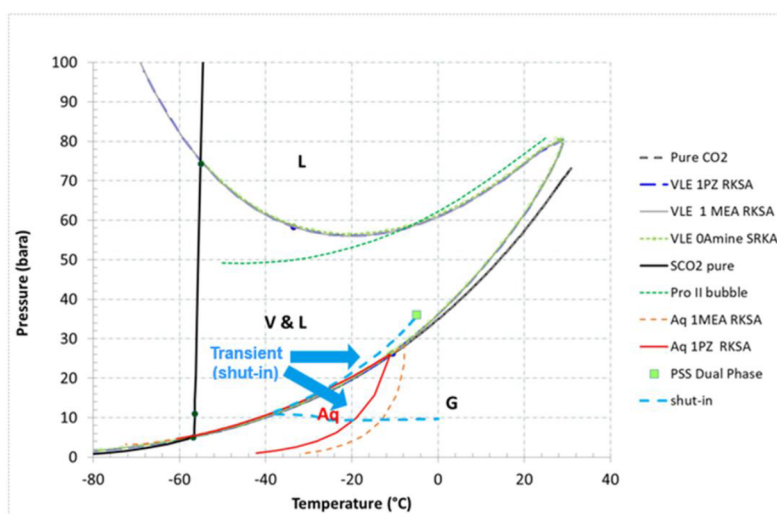


Figure 16—Transient from Dual-phase Pseudo Steady State excursion (blue dotted line) versus aqueous zone (in red)

Application of innovative design to the ARAMIS phase 1 project

Constraints

Many constraints were given by the project for the well completion design (initial design performed in 2024). Indeed, wells must be able to:

- Meet injection target rate (2.5 Mtpa) for 15 years and be able to inject up to 5 Mtpa during the first five years to back-up other operator in case of problem on their wells / platforms (contractual constraint).
- Have no gap in doable injection rates above 1 Mtpa to minimize start/stop of the wells or use extensive onshore buffering if CO₂ that can be injected differs from nominal target rate (operational constraint).
- Accommodate uncertainties linked to injectivity (Skin, Permeability), CO₂ composition (mix between cryogenic CO₂ and CO₂ coming from the industry), seasonal temperature variations (impacting the CO₂ wellhead injection temperature).
- Provide durability. Hence, completion will have to be suitable for early life (when reservoir is depleted and wells have too much injectivity) and late life (when reservoir is repressurized and injectivity must be maximized). The aim is to avoid costly workovers.
- Provide redundancy to injection target rate. That means that if one well is down, it is still possible to inject target rate thanks to the other wells.

Wells design

4 injectors will be used to inject the target rate. Figure 17 explains how it has been possible to accommodate all constraints given by the project using the innovative design. It is a snapshot at early life, for a winter case with zero skin (other injectivity and seasonality scenarios have been studied but are not presented). One has played on three elements:

- The first one is the variety of tubing size (top left plot). The four injectors are numbered 1, 2, 3 and 4 on the abscissa. All possible combinations of these 4 wells are also plotted. 12 represents the combination of well 1 and 2. 134 is the combination of wells 1, 3 and 4 etc. On the ordinate, one can read the doable injection rates considering only monophasic injection at the wellhead. The maximum rate is given by the well fully open and the minimum rate is given by the well choked down to 70b. Wells 1, 2 and 3 have similar injectivity whereas well 4 is expected to have

an injectivity 2 times better than the other wells. Well one will be equipped with a tubing 3 ½, well 2 with a tapered string 4 ½ × 3 ½ and wells 3 & 4 will be equipped with 4 ½ tubings. On that plot it can be seen that:

- Wells 3 & 4 (equipped with the biggest tubings) can make the target rate. Note that potential will decrease in summer so only well 4 will be able to inject target rate in summer.
- No combination of wells can make the target injection rate (too high potential).
- There are 2 gaps in doable injection rate between 1.7 and 2.2 Mtpa, and between 3.2 and 3.4 Mtpa (green squares on the plot).
- The second one is the ability to choke at the wellhead until -5°C and operate the well with dual phase injection (top right plot). Thanks to that:
 - On top of wells 3 and 4, injection rate can also be made with the combination of wells 1&2;
 - Gap between 1.71 and 2.15 Mtpa has been reduced and second gap has disappeared
- The last one is the ability to install retrievable DHCs in the wells (bottom right plot). 3 DHCs are installed on well 3, and 2 DHCs are installed on well 4. After installation of these DHCs:
 - Target injection rate can be made with 5 different combinations of 2 wells (full redundancy of the system).
 - There is no gap in doable injection rates.

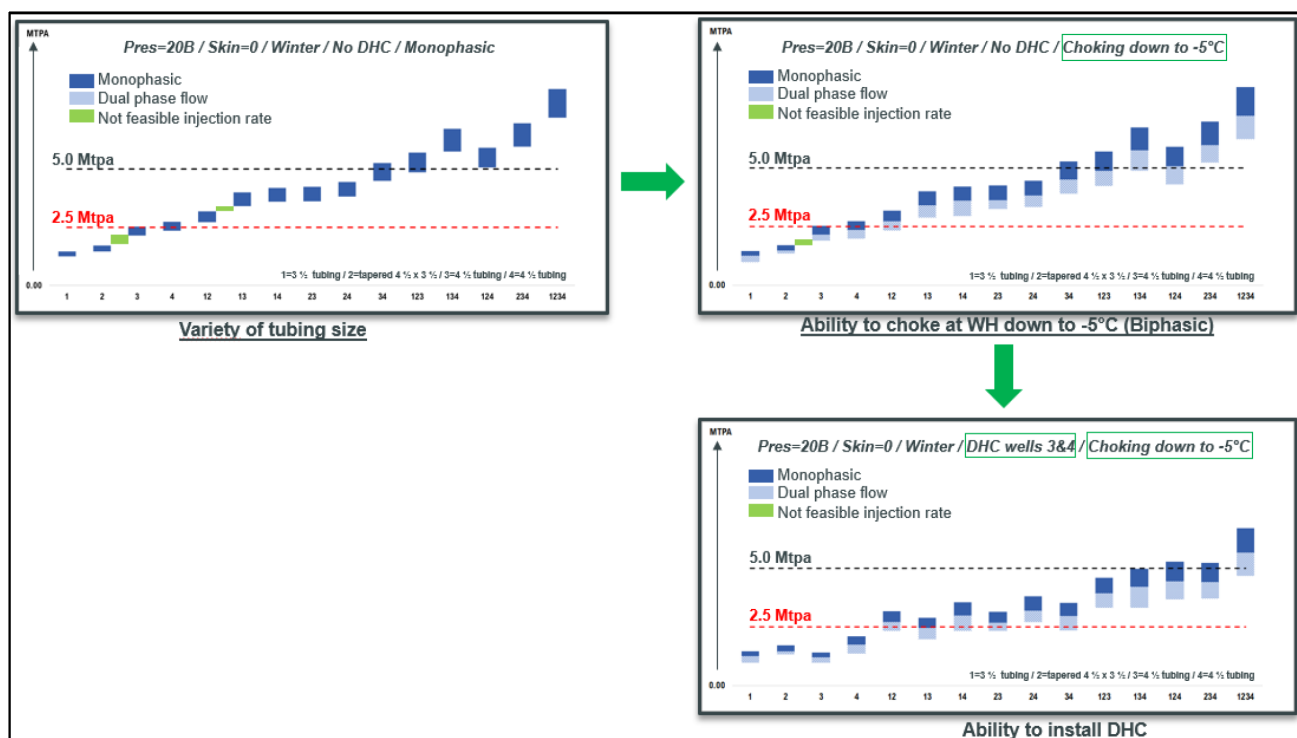


Figure 17—Wells design for ARAMIS Phase 1 based on innovative design and project constraints

Operating envelope versus time

Figure 18 show the overall well operating envelopes function of time (via the reservoir pressure increase) for the winter season with 0 skin case. In abscissa, one can read the injection rate and in ordinate the reservoir pressure. As such, start of injection corresponds to top of ordinates and end of injection corresponds to bottom of ordinates (as ordinate axis is reversed). On that plot, the combination of well operating envelopes

of one well are plotted in blue, the well operating envelopes with combinations of two wells are plotted in yellow, combinations of 3 wells are plotted in green and combinations of 4 wells are plotted in red.

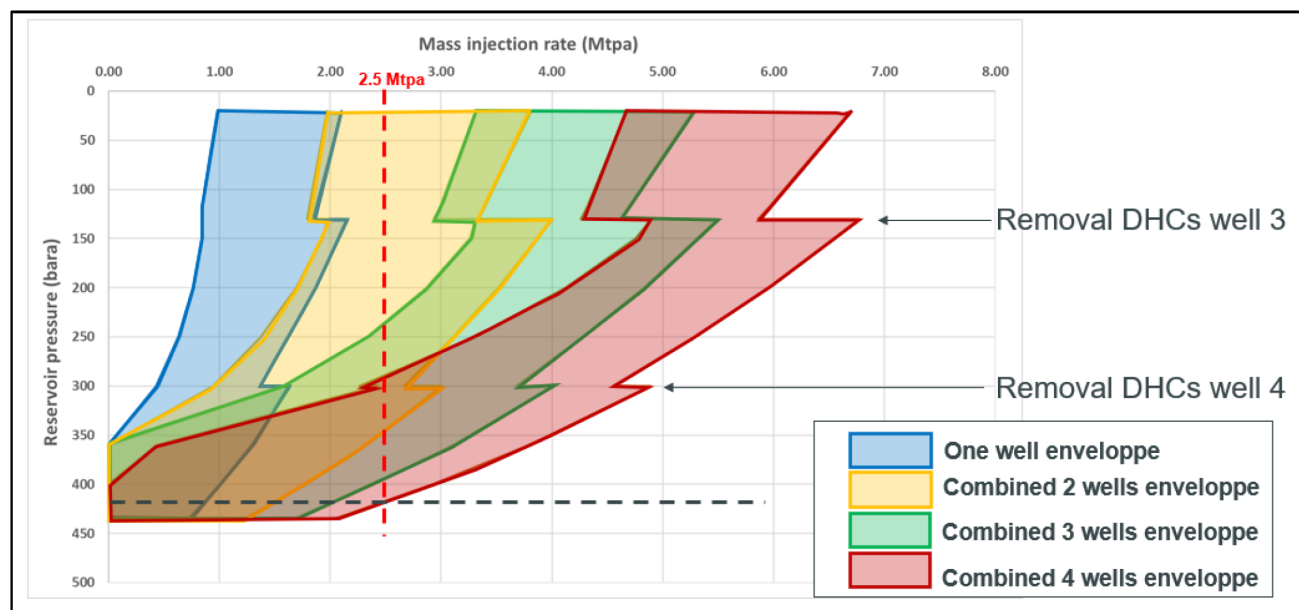


Figure 18—ARAMIS Phase 1 operating envelope versus time – Skin 0 – Winter Season

On that figure:

- The 2 bumps correspond to the removal of downhole chokes on wells 3 and 4 to gain some potential. The downhole chokes are removed on well 3 quite early due to the constraint of being able to inject 5 Mtpa during the first five years;
- The injection potential decreases when the reservoir repressurizes as expected;
- With retained completion design, target injection rate can be made until getting a reservoir pressure of 420b. At the end of 15 years, reservoir pressure is expected to be around 360b, so design is fine to make commercial contracts.

Additionally, the new design is expected to avoid performing an average of 4 costly workovers during the full injection lifespan, considering uncertainty on injectivity (in the low / mid / high cases injectivity, it is expected that 7 / 2 / 4 workovers would have been needed with traditional well design).

Conclusion

CO₂ storage in depleted gas reservoirs offers promising perspectives for decarbonation. Nonetheless, the injection of CO₂ in such depleted reservoirs comes with many challenges. Single phase injection offers a robust solution, however it lacks operational flexibility and adaptability (to uncertainty or to evolution of injection parameters with time)

The concept of operating CO₂ injector wells in dual-phase for depleted reservoir storage offers several advantages over single phase:

- Larger operating envelopes of injector wells, no gaps in injection rates possible
- Fewer wells are needed to cover a given range of rates, redundancy is improved
- Fewer well switches (switching from big injector to small injector due to variation on CO₂ rate supply), hence fewer transients

- Overall better operational flexibility than in a single-phase system from shore to reservoir

When coupled with the use of Down Hole Choke the concept further allows to:

- Manage injectivity uncertainties such as initial skin value or injectivity evolution in time (e.g formation damage)
- Manage variabilities such as seasonal changes of CO₂ (seabed temperature or composition for instance)
- Extend the life of wells, i.e. no workovers are needed to cope with the increasing reservoir pressure due to refill

Therefore leading to significant savings in terms of CAPEX, OPEX and increased up-time.

Yet, this concept comes with several points of attention (annular fluid freezing, corrosion, erosion, more complex simulations) that need to be scrutinized but that can be mitigated.

Down Hole Chokes retrievable on Slickline represent a simple, rugged and robust solution. However, further development, in particular of surface-controlled adjustable downhole choking (FCV) would certainly bring even more flexibility and operational benefits, less intervention costs and less exposure of well intervention personnel.

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Acronyms

CCS	Carbon Capture & Storage
CFD	Computational Fluid Dynamics
CRA	Corrosion Resistant Alloy
DHC	Down Hole Choke
Mtpa	Million Ton Per Annum
PDC	Polycrystalline Diamond Compound
PSS	Pseudo Steady State
OCTG	Oil Country Tubular Goods
WOE	Well Operating Envelope

REFERENCES

1. «"Aramis project description",» [En ligne]. Available: <https://www.aramis-ccs.com/news/co2-specifications-for-aramis-transport-infrastructure>.
2. A. H. Petitt and M. R. Konopczynski, Tendeka. SPE-209705-MS Using a CCS Simulator to Maintain Liquid CO₂ in the Completion. *SPE Eur*.
3. M. R. Konopczynski and A. H. Petitt, Tendeka. The Practical Design of a Carbon Storage Injection Well Completion Using In-Line Flow Control Devices to Maintain Injection Stability. SPE-210651-MS. SPE Asia Pacific Oil & Gas Conference and Ex.
4. API Spec 19ICV. Interval Control Valves (ICV), Standard by American Petroleum Institute, 06/01/2023.

5. Shane Turcott: (26) It is not the ice that causes pipe bursting | LinkedIn.
6. P. Mauger, P. Martin D. Rodriguez-Barrera Vallourec, S. Khoo, V. Bochkarev, H. Poitrenaud, M. Sofyan TotalEnergies. "Qualification Approach of OCTG Connections for CCS Injection Wells Tubing and Casing". OTC-35382-MS Offshore Technology Conference Houston, TX, U.
7. Martin, Pierre, et al "Tubing Connection Evaluation Methodology and Test Protocol for CCS Injection Wells." Proceedings of the *16th Greenhouse gas Control Technologies Conference (GHGT-16)*, 2022.
8. M. Sofyan, H. M. Poitrenaud, S. Khoo, A. Chilaka, B. Dechelette and E. V. Garcia, TotalEnergies. "Proposed Guidelines for Carbon Capture & Storage (CCS) Completions Design in Highly Depleted Reservoir". OTC-35200-MS. Offshore Technology Conference Houston.
9. M. Yazdanpanah, L. Galliot, S. Perrin, G. Moulie and S. Raurich. Review Of CO2 Specifications in Novel Industrial CCS Transportation and Storage Hubs. SPE-222240-MS. ADIPEC 2024.
10. C. Millet, G. Neel, L. Lima, D. Rordrigez Barrera, "Material Selection for anthropogenic CO2 injection: mechanical and corrosion performance of steels under dense CO2 with impurities" Paper 18886. AMPP annual conference 2023.